Chapter – 6: Materials Around Us

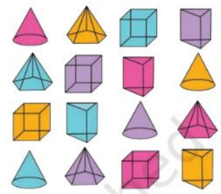
The materials used to make the crucible (a vessel used to melt substances) are clay and iron. – (Rasaratnasamuchchaya–10.3)

Observing Objects Around Us

- Ghulan, Sheeta – thrilled – new class – after summer break
- Enter classroom – start talking
- Sheeta – what have you brought today
- Ghulan – new notebook
- Sheeta – also new notebook, new pen
- After some time – madam Vidya – science teacher – enters classroom – interacts with students – things – used in daily life
- List of ques. –
    - Are they similar or diff.
    - Their shapes and colors
    - Feeling on touching them
    - Are some heavier than others
- All thing – made of something – paper, wood, cloth, glass, metal, plastic, clay, etc.
- Any substance – used to create obj. – **material**
- Activity –
    - Make a list – obj. around us
    - Also note – materials – they are made of
- Everyday obs. – obj. – made of various materials

How to Group Materials

- Activity –
    - Grp. obj. – shown here – common prop. – shape, color, hardness, softness, shine, dullness, materials
    - Discuss – which prop. you used – which prop. friends used
- Notice – obj. – made of diff. materials – OR – same material – diff. obj.
- Method of arranging obj. – in grp. – **classification**
- Obj. – classified – common prop.
- Activity –
    - Name materials – used to make a tumbler
- These materials – should hold water
- Choose a material – make an obj. – depends on – prop. of material / purpose of obj.
- May use – diff. materials – diff. parts of obj.
- Ex. – pen – made of – plastic, metal, ink
- Activity –
    - Take some balls – same size – diff. materials
    - Drop each ball – some height



- Note – height – ball bounces upto – **record it**
- Identify – ball – highest bounce

- Discuss in class – other prop. – sports balls – size, color, texture, height of bounce
- Obs. fig. – grp. obj. – as many ways as possible
- These obj. – grp. – based on – shape, color, material
- We learnt – materials – classified – based on prop.
- Ex. – in kitchen – store things – similar utensils – placed together
- Similarly – grocer – keeps all spices – together – pulses, grains together BUT diff. place
- Visit chemist – enquire – medicines arrangement

What are Different Properties of Materials

Observe and identify appearance of materials

- Materials – look diff. – each other
- Freshly cut wood – unpolished – distinct appearance – very diff. – iron
- Similarly – iron – diff. from – copper, aluminium
- BUT – some similarities – iron, copper, aluminium – diff. from wood
- Sorting challenge –
    - Collect – paper, cardboard, chalk, wood, copper wire, aluminium foil, any article – made of brass, bronze, steel, etc.
    - Take closer look – these materials – do they shine under light
    - Obs. their texture – rough or smooth – color – other noticeable features
    - Record obs. – grp. – based on appearance
- Materials – shiny surfaces – **lustrous**
- Materials with lustre – usually metals – iron, copper, zinc, aluminium, gold, etc.
- Some metals – loose lustre – look dull, non-lustrous – effect of air, moisture
- Result – often notice – lustre – only on freshly cut surface
- **Non-lustrous** materials – do not have – shiny surface
- Ex. – paper, wood, rubber, juice, etc.
- ‘All that glitters – not gold’ – old saying
- All materials – shine – not metals
- Some materials – made shiny – polishing, coating layers – plastic, wax – may not be metals

Which materials are hard

- Press diff. obj. – your hand – some – stone – hard to compress – others – eraser – easily compressed
- Take metal key – scratch surface – wood, aluminium, stone, iron, candle, chalk, other obj.
- Materials – compressed, scratched easily – **soft**
- Materials – difficult to compress, scratch – **hard**
- These prop. – rel. in nature – rubber – harder than sponge – softer than iron
- Activity –
    - Take some obj. – brick, water bottle, pillow, tumbler, table, sweater
    - Feel them – soft or hard – what material – note obs.
    - Compare with friends
- Materials – diff. prop. – lustre, hardness, softness, colour

Materials through which one can see or cannot see

- Ghulan, Sheeta, Sara – hide and seek
- Ghulan – behind a wall – Sheeta – behind a tree – Sara – behind a frosted glass door
- Sheeta's brother – obs. everything – glass window
- Materials – through which one can see easily – **transparent**
- Ex. – glass, water, air, cellophane paper, etc.
- Many materials – through which one cannot see at all – **opaque**
- Ex. – wood, cardboard, metals
- Some materials – one can see but not clearly – **translucent**
- Ex. – butter paper, frosted glass

What is soluble in water; what is not

- Ghulan – sweating – after playing – feeling tired, thirsty
- His mother – mixed – spoonful of sugar, pinch of salt, some lemon juice, glass of water – offered *shikanji* (lemonade)
- He noticed – sugar, salt – disappeared in water
- Activity –
    - Collect small amt. – sugar, salt, chalk powder, sand, sawdust
    - Take 5 tumblers – fill them – 2/3rd water
    - Put spoonful of each item in separate tumbler
    - **Predict** – what will happen
    - Stir them – wait few min. – note obs.
- Notice – some materials – dissolve completely – **soluble**
- Other materials – do not dissolve OR disappear – stir for long – **insoluble**
- Water – imp. role – func. – our body – dissolve lots of materials
- Some liquids – dissolve completely – others – form separate layer
- Similarly – some gases – soluble – others – not soluble
- Ex. – O_2 – dissolves in water – very imp. – survival of plants and animals – lives in water

How heavy or light

- Activity –
    - Take 3 identical paper cups – fill them half – provided materials
    - 1st cup – water – mark as A
    - 2nd cup – sand – mark as B
    - 3rd cup – pebbles – mark as C
    - Predict – heavier or lighter material
    - Weigh each cup – use a balance – record obs.
- Any obj. – heavier or lighter – measured in terms of prop. – **mass**
- Heavier – more mass – lighter – less mass

Space and volume

- Next day – madam Vidya – enters the class
- All students – stand up – greet her
- She replies – keep your bags on seat – sit down
- Students – not able to sit – she asks again – why not seating
- Students reply – bags already occupy space
- Now – she provides – 2 identical glass tumblers

- Encourage students – pour water – water bottle to tumbler
- Students obs. – one tumbler – gets half filled – other – completely filled
- Madam Vidya – explains – both tumblers – same capacity
- Diff. water level – indicate – amt. of water – each case – diff.
- Water – 1st tumbler – less space – indicate – vol. of water – lesser than other tumbler
- Space – occupied by water – **volume**
- Water bottles – diff. sizes – sold in market – 1 L, 500mL, 300 mL, etc. – net qty.

- Now – familiar with many prop. – BUT – all materials – do not have – all prop.

What is Matter?

- Mass, vol. – 2 prop. – all materials
- Anything – occupy space – contain mass – **matter**
- Mass – qty. of matter – units to measure – gram (g), kilogram (kg)
- Vol. – space occupied – units to measure – litre (L), millilitre (mL)
- All materials – around us – ex. of matter – water, pebbles, sand, cup, etc.
- Materials – types of matter – create (make) obj.
- Materials – look diff. – behave diff.
- Grp. materials – based on similarities, diff. – their prop.
- Grouping – useful – helps – study, obs. – patterns
- Humans – classify everything – living – plants, animals – OR – non-living – rocks, other obj.